

Hyperbolic trig identities

Key Points

The identities to remember are

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\cosh^2 x - \sinh^2 x = 1$$

$$\operatorname{sech}^2 x + \tanh^2 x = 1$$

$$\coth^2 x - \operatorname{cosech}^2 x = 1$$

$$\sinh 2x = 2 \sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x$$

Basic Skills

Q1. Write down the exponential forms of

a) $y = \cosh x$

b) $y = \operatorname{cosech} x$

c) $y = \sinh x$

d) $y = \operatorname{sech} x$

e) $y = \tanh x$

f) $y = \coth x$

Q2. Which of the identities is really a definition

Q3. Prove the identity $\cosh^2 x - \sinh^2 x = 1$ and explain the geometric relevance of the equation

Q4. Solve $\sinh^2 x - 2 \sinh x - 3 = 0$

Competency Skills

Q5. Prove the identity $\operatorname{sech}^2 x + \tanh^2 x = 1$ from exponential form

Q6. Prove the identity $\coth^2 x - \operatorname{cosech}^2 x = 1$ by using the identity $\cosh^2 x - \sinh^2 x = 1$

Q7. Solve the following equations

a) $\cosh 2x - 8 \sinh x = 3$

b) $\sinh^2 x = 5 \cosh x + 2$

c) $3 \operatorname{sech}^2 x = \tanh x$

d) $\coth^2 x - \frac{2}{\sinh x} = 2$

Q8. Find an expression for $\tanh 2x$ in terms of $\tanh x$

Q9. Prove the identity $\cosh 3x = 4 \cosh^3 x - 3 \cosh x$

Advanced Skills

Q10. Find an expression for $\sinh 3x$

Q11. Prove the addition identity $\sinh(A + B) = \sinh A \cosh B + \cosh A \sinh B$

Q12. For the equation $a \sinh x - b \cosh x = c$ for constants a, b, c

a) Explain why the equation cannot be solved using the common identities

b) Using the exponential form solve find the solution in terms of a, b, c

c) Using $\sinh(A + B) = \sinh A \cosh B + \cosh A \sinh B$ solve the equation again

Extension

Q13. Explain why proving multiple argument identities is easy with hyperbolic trig than standard trig